

1) Step 1: Start ROS environment

```
source /opt/ros/noetic/setup.bash
source ~/catkin_ws/devel/setup.bash
roscore
```

2) Step 2: Start Care-O-Bot simulator

```
source /opt/ros/noetic/setup.bash
source ~/catkin_ws/devel/setup.bash
export ROBOT=cob4-5
export ROBOT_ENV=ipa-Nirish
roslaunch cob_bringup_sim robot.launch
```

3) Step 3: Check the camera

```
source /opt/ros/noetic/setup.bash
source ~/catkin_ws/devel/setup.bash
roslaunch image_view image_view image:=/head_cam/image_color
```

4) Step 4: Start SLAM mapping

```
source /opt/ros/noetic/setup.bash
source ~/catkin_ws/devel/setup.bash

roslaunch gmapping slam_gmapping \
scan:=/scan_unified \
_base_frame:=base_link \
_odom_frame:=odom_combined
```

5) Step 5: Move the robot

```
source /opt/ros/noetic/setup.bash
source ~/catkin_ws/devel/setup.bash

roslaunch teleop_twist_keyboard teleop_twist_keyboard.py \
cmd_vel:=/base/twist_mux/command_navigation
```

6) Step 6: Open RViz

```
source /opt/ros/noetic/setup.bash
source ~/catkin_ws/devel/setup.bash
rviz
```

Add

```
|  
|— Map  
|— TF  
|— RobotModel  
|— LaserScan (Topic: /scan_unified)
```

7) Step 7: YOLO detector

```
source /opt/ros/noetic/setup.bash  
source ~/catkin_ws/devel/setup.bash  
roslaunch privacy_environment_map privacy_detector.py _camera_id:=left  
_node_name:=privacy_detector_left  
roslaunch privacy_environment_map privacy_detector.py _camera_id:=right  
_node_name:=privacy_detector_right
```

```
source ~/catkin_ws/devel/setup.bash  
roslaunch privacy_environment_map privacy_detector.py  
rostopic echo /privacy_detections
```

```
privacy_environment_map  
|  
|— scripts  
|   |  
|   |— privacy_detector.py  
|  
|— models  
|   |  
|   |— best.pt
```

```
cd ~/catkin_ws/src/privacy_environment_map  
chmod +x scripts/privacy_detector.py  
cd ~/catkin_ws  
catkin_make  
source devel/setup.bash  
roslaunch privacy_environment_map privacy_detector.py
```

8) Step 8: Create object position node using Depth camera

```
source /opt/ros/noetic/setup.bash  
source ~/catkin_ws/devel/setup.bash  
roslaunch privacy_environment_map object_position.py _camera_id:=left  
__name:=object_position_left
```

```
roslaunch privacy_environment_map object_position.py _camera_id:=right  
__name:=object_position_right
```

```
source ~/catkin_ws/devel/setup.bash  
roslaunch privacy_environment_map object_position.py  
/privacy_objects_map
```

```
cd ~/catkin_ws/src/privacy_environment_map  
chmod +x scripts/object_position.py  
cd ~/catkin_ws  
catkin_make  
source devel/setup.bash  
roslaunch privacy_environment_map object_position.py
```

9) Publish privacy markers

```
source /opt/ros/noetic/setup.bash  
source ~/catkin_ws/devel/setup.bash  
roslaunch privacy_environment_map privacy_marker.py _camera_id:=left  
__name:=privacy_marker_left  
roslaunch privacy_environment_map privacy_marker.py _camera_id:=right  
__name:=privacy_marker_right
```

```
source ~/catkin_ws/devel/setup.bash  
roslaunch privacy_environment_map privacy_marker.py  
rostopic echo /privacy_markers
```

```
cd ~/catkin_ws/src/privacy_environment_map  
chmod +x scripts/privacy_marker.py  
cd ~/catkin_ws  
catkin_make  
source devel/setup.bash  
roslaunch privacy_environment_map privacy_marker.py
```

10) Step 10: Add Privacy Marker in RViz

Add:

Marker (Topic: /privacy_markers)